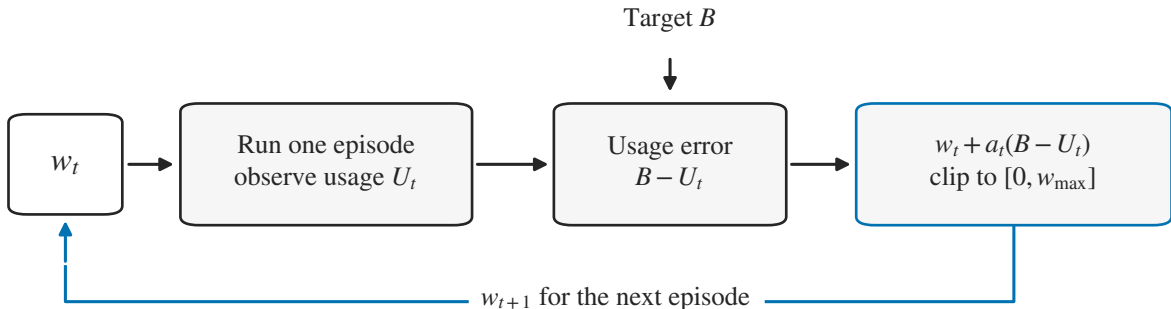




$U_t < B$ pushes w up

$U_t > B$ pushes w down

Stationary guarantee requires a single sign-consistent crossing and the stated assumptions.

Decaying step $a_t = \eta_0 / (1 + \delta t)$, $\delta > 0$. Reset-on-shift is a heuristic.